

THE RATIONAL HODGE CONJECTURE FOR ALL POWERS OF CYCLIC CUBIC FOURFOLDS

GRANT STENGER

ABSTRACT. Let X be a smooth complex cubic fourfold. We prove that if the transcendental Hodge endomorphism field of X is CM, then every rational Hodge class on every self-power of X is algebraic. The two principal ingredients are a codimension-four Chow realization of the entire endomorphism field, obtained through the Fano variety of lines and Markman’s algebraicity theorem, and a rational unitary invariant calculation showing that these two-factor classes generate all tensor invariants. The deck transformation of a smooth cyclic cubic embeds $\mathbf{Q}(\zeta_3)$ in its transcendental endomorphism field, so the result applies to every smooth member of the cyclic family. We also construct an explicit Pfaffian-type cycle spanning the primitive determinant line and apply published split Chow-motive retractions to associated Kuznetsov-component moduli spaces and OG10 resolutions.

1. INTRODUCTION AND MAIN RESULTS

The rational Hodge conjecture is known for a smooth complex cubic fourfold itself, by work of Zucker [25]. This does not formally control its self-products. The cohomology of X^m contains tensor powers of the transcendental primitive cohomology of X , and their invariant tensors produce Hodge classes that have no counterpart on a single factor. Thus an all-power theorem requires two additional ingredients: a description of all tensor invariants and algebraic correspondences realizing a set of generators for them.

This paper gives such a description when the transcendental Hodge endomorphism field is CM. In that case Zarhin’s Hodge group is unitary. After extension of scalars, its tensor invariants are generated by complete pair contractions. The corresponding rational two-factor tensors are the Hodge endomorphisms of the transcendental cohomology. The geometric problem therefore collapses to algebraizing one finite-dimensional field on $X \times X$.

The Fano variety of lines provides the geometric bridge. Norm-one elements span the CM field, and each norm-one element is a Hodge isometry. Through the Beauville–Donagi Abel–Jacobi isomorphism it extends by the identity on the Néron–Severi group to a rational Hodge isometry of the full second cohomology of $F(X)$. Markman’s theorem makes that isometry algebraic. A g^2 -corrected incidence correspondence transports it back to the cubic. The complete proof chain is

$$\begin{aligned} E_X \text{ CM} &\implies E_X = \langle U_{E_X}(\mathbf{Q}) \rangle_{\mathbf{Q}} \\ &\implies U_{E_X}(\mathbf{Q}) \text{ is algebraic through } F(X) \\ &\implies E_X \subset \text{cl}(\text{CH}^4(X \times X)_{\mathbf{Q}}) \\ &\implies (T_X^{\otimes r}(2r))^{\text{Hg}(T_X)} \text{ is algebraically generated by two-factor tensors} \\ &\implies \text{HC}_{\mathbf{Q}}(X^m) \quad \text{for every } m \geq 1. \end{aligned}$$

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Two transitions in this chain are essential: the passage from algebraicity through $F(X)$ to Chow correspondences on $X \times X$, and the passage from those correspondences to all invariant tensors. The former is geometric: the g^2 -weighted incidence return converts the algebraic isometries on $F(X)$ into rational Chow correspondences on $X \times X$, rather than merely Hodge endomorphisms. The latter is representation-theoretic: because the Hodge group is the full unitary group, every invariant is generated by pair contractions, and no independent special-unitary determinant tensor remains.

Main results. Let X be a smooth cubic fourfold. Put

$$V_X = H_{\text{prim}}^4(X, \mathbf{Q}), \quad Q_X(x, y) = \int_X x \cup y,$$

and set

$$N_X = V_X \cap H^{2,2}(X), \quad T_X = N_X^{\perp, Q_X} \subset V_X, \quad E_X = \text{End}_{\text{HS}}(T_X).$$

Thus N_X is the space of rational primitive Hodge classes and T_X is its orthogonal complement for the primitive intersection form. We emphasize that the hypothesis “ E_X is CM” means that the *endomorphism field* is a CM number field; it does not mean that T_X is a CM Hodge structure or that its Hodge group is a torus. All Chow groups below have rational coefficients and all varieties and cycles are over $\mathbf{C}$. For a smooth projective variety Z , the notation $\text{HC}_{\mathbf{Q}}(Z)$ means that the rational cycle class map is surjective onto the rational Hodge classes in every codimension.

Theorem A (CM algebraization and all powers). *If E_X is a CM field, then every element of E_X is represented by an algebraic correspondence: for every $e \in E_X$ there is $\Gamma_e \in \text{CH}^4(X \times X)_{\mathbf{Q}}$ such that $(\Gamma_e)_*|_{T_X} = e$. Moreover,*

$$\text{HC}_{\mathbf{Q}}(X^m)$$

holds in every codimension for every $m \geq 1$.

We next specialize to the cyclic family. Let F define a smooth cubic threefold and put

$$Y = \{x_5^3 + F(x_0, \dots, x_4) = 0\} \subset \mathbf{P}^5, \quad \tau[x_0 : \dots : x_5] = [x_0 : \dots : \zeta_3 x_5].$$

Write $h \in \text{CH}^1(Y)$ for the hyperplane class and

$$V = H_{\text{prim}}^4(Y, \mathbf{Q}), \quad Q_Y(x, y) = \int_Y x \cup y, \quad \dim_{\mathbf{Q}} V = 22.$$

The algebraic primitive projector is

$$\pi_V = \Delta_Y - \frac{1}{3} \sum_{i=0}^4 h^{4-i} \times h^i \in \text{CH}^4(Y \times Y)_{\mathbf{Q}}. \quad (1.1)$$

We compose right-to-left: $\Gamma_2 \circ \Gamma_1$ acts first by Γ_1 and then by Γ_2 . Set

$$\beta = \pi_V \circ (\Gamma_{\tau} - \Gamma_{\tau^{-1}}) \circ \pi_V \in \text{CH}^4(Y \times Y)_{\mathbf{Q}}. \quad (1.2)$$

For a permutation $\sigma \in \mathfrak{S}_{22}$, let the same symbol denote the automorphism of Y^{22} that permutes factors. Define the normalized alternator

$$P_{\text{alt}} = \frac{1}{22!} \sum_{\sigma \in \mathfrak{S}_{22}} \text{sgn}(\sigma) \sigma_*.$$

Theorem B (Cyclic application and graph-Pfaffian orientation). *Every smooth cyclic cubic fourfold Y has CM transcendental Hodge endomorphism field. Consequently*

$$\mathrm{HC}_{\mathbf{Q}}(Y^m)$$

holds in every codimension for every $m \geq 1$. In addition, the explicit cycle

$$z_\tau = P_{\mathrm{alt}}(\beta^{\boxtimes 11}) \in \mathrm{CH}^{44}(Y^{22})_{\mathbf{Q}} \quad (1.3)$$

where the eleven copies of β occupy the ordered pairs $(1, 2), (3, 4), \dots, (21, 22)$, has nonzero Betti class. More precisely,

$$0 \neq \mathrm{cl}(z_\tau) \in \bigwedge_{\mathbf{Q}}^{22} V \subset H^{88}(Y^{22}, \mathbf{Q}),$$

so it spans the rational primitive determinant line.

Recall that the Kuznetsov component is

$$\mathrm{Ku}(X) = \langle \mathcal{O}_X, \mathcal{O}_X(1), \mathcal{O}_X(2) \rangle^\perp \subset \mathrm{D}^b(X).$$

We write $\mathrm{Stab}^\dagger(\mathrm{Ku}(X))$ for its distinguished connected component of Bridgeland stability conditions, and use OG10 for O’Grady’s ten-dimensional deformation type.

Theorem C (K3-category moduli export). *Let X be a smooth cubic fourfold whose transcendental Hodge endomorphism field is CM; in particular, X may be any smooth cyclic cubic. Form a finite product from the following factors:*

- (i) *arbitrary powers of X ;*
- (ii) *nonempty smooth projective moduli spaces $M_\sigma(\mathrm{Ku}(X), v)$ of dimension $2m$, where v is primitive and $\sigma \in \mathrm{Stab}^\dagger(\mathrm{Ku}(X))$ is v -generic;*
- (iii) *any projective hyperkähler cubic-Kuznetsov OG10 crepant resolution in the scope of Fu–Laterveer–Vial’s theorem, arising from $v = 2v_0$ with v_0 primitive and $v_0^2 = 2$, and with $\sigma \in \mathrm{Stab}^\dagger(\mathrm{Ku}(X))$ v -generic, including the projective symplectic resolution constructed by Li–Pertusi–Zhao when its hypotheses hold; and*
- (iv) *smooth projective irreducible-symplectic varieties birational to any factor in (ii) or (iii). In the standard algebraic A_2 -lattice notation, when $v_0 = \lambda_1 + \lambda_2$, this includes the smooth projective twisted-intermediate-Jacobian compactification covered by the Li–Pertusi–Zhao birational comparison.*

Then the rational Hodge conjecture holds for that product in every codimension. There is no a priori bound on its dimension or number of factors.

Proof strategy. The argument separates into four steps.

1. *Algebraize the CM field.* The rational points of the norm-one torus span E_X . Each norm-one element extends to the full $H^2(F(X), \mathbf{Q})$ and is algebraized there by Markman’s theorem. The raw incidence maps have the wrong degree to return directly to X ; inserting g^2 , the square of the Plücker class, gives a codimension-four correspondence on $X \times X$ acting as the prescribed endomorphism of T_X .
2. *Generate every tensor invariant.* Zarhin’s description of the Hodge group and the first fundamental theorem for general linear groups show, after rational descent, that every tensor invariant is generated by these two-factor endomorphism tensors.
3. *Recombine the Künneth pieces.* Algebraic projectors separate transcendental, algebraic primitive, and ambient cohomology on every factor. Pullbacks and products of the endomorphism correspondences then give rational Chow representatives for all Hodge classes on X^m , proving Theorem A.

4. *Specialize and export.* The order-three deck transformation supplies a CM subfield for every smooth cyclic cubic and gives the nondegenerate alternating tensor whose top Pfaffian is z_τ . Tensoring published split Chow-motive retractions then proves Theorem C.

Relation to previous work. The theorem for one cubic is due to Zucker [25]. Shioda proved the Hodge conjecture for the Fermat cubic and all of its powers [19, Theorem 2], so Theorem A is not the first all-power result for a cubic. For cyclic cubics, van Geemen–Izadi constructed algebraic half-twist and Kuga–Satake correspondences [20, Theorem 5.2 and Corollary 5.3], and Laterveer proved a finite-dimensional motive of abelian type statement [11, Theorem 3.1]. Schoen’s 1988 construction for self-products endowed with an automorphism is an important precedent [17]. Neither finite-dimensionality nor membership in the tensor category generated by abelian motives, by itself, supplies algebraic representatives for every Hodge class on every power. The additional step here is to exhibit enough Chow correspondences on the cubic to generate the full tensor-invariant algebra.

The CM square-to-all-powers mechanism has a direct K3 precedent. Ramón Marí proved that, for a K3 surface with CM transcendental Hodge endomorphism field, the tensor Hodge algebra is generated by classes on the square and concluded the Hodge conjecture for every self-power [15, Proposition 5.2 and Theorem 5.4]. Varesco gives a modern proof of the CM invariant statement [21, Theorem 2.2] and identifies the error in the totally real extension asserted in Ramón Marí’s Proposition 5.2: special-orthogonal determinant tensors supply additional generators [21, Remark 2.17]. Only the unaffected CM branch is used here.

The geometric input in the present paper is the algebraicity on a cubic fourfold of the entire transcendental endomorphism field. Markman’s theorem algebraizes rational second-cohomology Hodge isometries of the Fano variety of lines [13, Theorem 1.1]; the full- H^2 extension and the correctly typed g^2 -weighted incidence return then produce codimension-four Chow correspondences on the cubic. Section 3 reproduces the CM invariant argument for the polarized K3-type Hodge structure $T_X(1)$ and includes the rational descent. The Pfaffian-type graph cycle is a separate explicit refinement for cyclic cubics, and the K3-category consequences use the split Chow-motive theorems of Fu–Laterveer–Vial and Floccari–Fu–Zhang.

Conventions and scope. All varieties and cycles are over $\mathbf{C}$, and all Chow groups and motives have rational coefficients. We compose correspondences in the order in which they act. If a cyclic equation is defined over $\mathbf{Q}$, Galois conjugation may exchange τ and τ^{-1} and negate z_τ ; arithmetic descent is not asserted. Theorem A does not cover totally real endomorphism fields. The results do not concern integral Hodge classes, products attached to different cubics, moduli spaces outside the stated split-motive scope, or the general Hodge conjecture.

The paper is organized as follows. Section 2 constructs the Chow projectors and algebraizes the CM endomorphism field through the Fano variety. Section 3 proves rational unitary invariant generation and Theorem A. Section 4 establishes the cyclic application and explicit graph-Pfaffian. Section 5 proves the K3-category export, and Section 6 records the precise scope of the method. The supplementary generic Hodge-group calculation in the appendix is not used in the proof of any main theorem.

2. HODGE STRUCTURES AND CM ALGEBRAIZATION THROUGH THE FANO VARIETY

Let X be a smooth cubic fourfold, let $F_X = F(X)$ be its Fano variety of lines, and let $P \subset F_X \times X$ be the universal line. Write

$$A = {}^tP \in \mathrm{CH}^3(X \times F_X)_{\mathbf{Q}}.$$

We will repeatedly use the following variance convention. If U is smooth projective of dimension d and $\Gamma \in \mathrm{CH}^r(U \times V)_{\mathbf{Q}}$, then

$$\Gamma_* : H^i(U, \mathbf{Q}) \longrightarrow H^{i+2(r-d)}(V, \mathbf{Q}). \quad (2.1)$$

If $\Gamma_1 \in \mathrm{CH}^r(U \times V)$ and $\Gamma_2 \in \mathrm{CH}^s(V \times W)$, then $\Gamma_2 \circ \Gamma_1$ has codimension $r + s - \dim V$. Thus A_* sends $H^4(X)$ to $H^2(F_X)$; this numerical check will make the two degree corrections below transparent. For an automorphism f of a smooth projective variety, Γ_f denotes its graph; with this action convention $(\Gamma_f)_* = f_* = (f^{-1})^*$.

2.1. Chow projectors on a cubic fourfold. Write $h \in \mathrm{CH}^1(X)$ for the hyperplane class and put

$$\pi_i^{\mathrm{amb}} = \frac{1}{3} h^{4-i} \times h^i, \quad \pi_{\mathrm{amb}} = \sum_{i=0}^4 \pi_i^{\mathrm{amb}}, \quad \pi_V^X = \Delta_X - \pi_{\mathrm{amb}}. \quad (2.2)$$

Thus π_V^X is the primitive middle projector. By Zucker's theorem, choose cycles $a_1, \dots, a_s \in \mathrm{CH}^2(X)_{\mathbf{Q}}$ whose classes form a basis of N_X , and let $G = (Q_X(a_i, a_j))$. Define

$$\pi_N^X = \sum_{i,j} (G^{-1})_{ij} a_i \times a_j, \quad \pi_T^X = \pi_V^X - \pi_N^X, \quad \pi_{\mathrm{alg}}^X = \Delta_X - \pi_T^X. \quad (2.3)$$

Lemma 2.1 (Chow idempotents). *The four correspondences π_{amb} , π_V^X , π_N^X , and π_T^X are idempotent in $\mathrm{CH}^4(X \times X)_{\mathbf{Q}}$. The first two pairs in the decomposition*

$$\Delta_X = \pi_{\mathrm{amb}} + \pi_N^X + \pi_T^X$$

are mutually orthogonal, and their Betti realizations project respectively onto the ambient cohomology, N_X , and T_X . Consequently π_{alg}^X is a Chow idempotent whose realization is the graded algebraic complement of T_X in $H^(X, \mathbf{Q})$.*

Proof. Since $\int_X h^4 = 3$, direct composition of decomposable correspondences shows that the five π_i^{amb} are mutually orthogonal idempotents. Their realizations project onto the five ambient lines $\mathbf{Q}h^i$, $0 \leq i \leq 4$, so $\pi_V^X = \Delta_X - \pi_{\mathrm{amb}}$ is the primitive middle idempotent.

A smooth cubic fourfold is Fano and hence rationally connected; therefore $\mathrm{CH}_0(X)_{\mathbf{Q}} = \mathbf{Q}$. With the degree-one cycle $o_X = h^4/3$, one has in $\mathrm{CH}_0(X)_{\mathbf{Q}}$

$$a_i \cdot a_j = G_{ij} o_X. \quad (2.4)$$

For decomposable middle correspondences,

$$(a_i \times a_j) \circ (a_k \times a_l) = G_{li} a_k \times a_j.$$

Writing $M = G^{-1}$, matrix multiplication gives explicitly

$$(\pi_N^X)^2 = \sum_{i,j,k,l} M_{ij} M_{kl} G_{li} a_k \times a_j = \sum_{k,j} M_{kj} a_k \times a_j = \pi_N^X.$$

Thus idempotency holds in Chow, not only after realization. Each a_i is primitive, so $\int_X a_i h^2 = 0$; the same composition formula gives $\pi_{\mathrm{amb}} \pi_N^X = \pi_N^X \pi_{\mathrm{amb}} = 0$, and hence $\pi_V^X \pi_N^X = \pi_N^X \pi_V^X = \pi_N^X$. It follows that

$$(\pi_T^X)^2 = (\pi_V^X - \pi_N^X)^2 = \pi_T^X$$

and that the displayed decomposition is orthogonal. The assertions about the Betti images follow from the definitions and the nonsingularity of G . $\square$

2.2. The transcendental summand of the Fano variety. Let

$$T_{F_X} = \mathrm{NS}(F_X)_{\mathbf{Q}}^{\perp, q_{F_X}} \subset H^2(F_X, \mathbf{Q})$$

be the transcendental second cohomology, where q_{F_X} is the Beauville–Bogomolov form. The incidence Abel–Jacobi map is an isomorphism on primitive cohomology. We record the exact transcendental consequence.

Lemma 2.2 (Abel–Jacobi and the transcendental decomposition). *The incidence correspondence induces a polarized Hodge isomorphism*

$$A_* : V_X(1) \xrightarrow{\sim} g^{\perp} \subset H^2(F_X, \mathbf{Q}), \quad q_{F_X}(A_*x, A_*y) = -Q_X(x, y).$$

Moreover,

$$\mathrm{NS}(F_X)_{\mathbf{Q}} = \mathbf{Q}g \oplus A_*(N_X(1)), \quad T_{F_X} = A_*(T_X(1)). \quad (2.5)$$

Proof. Beauville–Donagi prove that A_* identifies the primitive middle cohomology of X , after the Tate twist, with the primitive second cohomology $g^{\perp}$ of F_X ; in the normalization used here it is the displayed anti-isometry [4, Proposition 6]; see also [10, Section 2.1 and Proposition 2.1.1]. The class g is of type $(1, 1)$, and A_* preserves Hodge type after the twist. Hence

$$\mathbf{Q}g \oplus A_*(N_X(1)) \subseteq \mathrm{NS}(F_X)_{\mathbf{Q}}$$

by the Lefschetz $(1, 1)$ theorem.

Conversely, take $\xi \in \mathrm{NS}(F_X)_{\mathbf{Q}}$ and decompose it orthogonally as $\xi = cg + \xi_0$, with $\xi_0 \in g^{\perp}$. The Beauville–Donagi isomorphism writes $\xi_0 = A_*x$ for a unique $x \in V_X(1)$. Since ξ_0 has type $(1, 1)$, the class x has type $(1, 1)$ after twisting, equivalently its untwisted class lies in $V_X \cap H^{2,2}(X) = N_X$. This proves the first equality in (2.5). Taking orthogonal complements and using the anti-isometry proves the second. $\square$

Lemma 2.3 (Simplicity of the transcendental structure). *The polarized Hodge structure $T_X(1)$ is simple. Consequently $E_X = \mathrm{End}_{\mathrm{HS}}(T_X)$ is a number field.*

Proof. The Hodge structure $T_X(1)$ is of K3 type, with a unique $(2, 0)$ line and no nonzero rational class of type $(1, 1)$. Let $W \subseteq T_X(1)$ be a nonzero rational Hodge substructure. If $W_{\mathbf{C}}$ misses the $(2, 0)$ line, then it also misses its conjugate $(0, 2)$ line and is therefore of type $(1, 1)$, a contradiction. If $W_{\mathbf{C}}$ contains the $(2, 0)$ line and W is proper, then its polarized orthogonal complement is a nonzero rational Hodge substructure of type $(1, 1)$, again a contradiction. Hence $W = T_X(1)$. The endomorphism theorem for simple K3-type Hodge structures then identifies E_X with a number field [24, Theorem 1.5.1]. $\square$

Lemma 2.4 (Norm-one spanning). *Let E/E_0 be a CM field and put*

$$U_E(\mathbf{Q}) = \{u \in E^{\times} : u\bar{u} = 1\}.$$

Then the $\mathbf{Q}$ -linear span of $U_E(\mathbf{Q})$ is all of E .

Proof. Hilbert theorem 90 makes the rational map

$$\mathrm{Res}_{E/\mathbf{Q}} \mathbf{G}_m \dashrightarrow \mathrm{Res}_{E_0/\mathbf{Q}} \mathrm{U}(1), \quad x \longmapsto x/\bar{x},$$

dominant. Since $E^{\times}$ is Zariski dense in $\mathrm{Res}_{E/\mathbf{Q}} \mathbf{G}_m$, the rational points $U_E(\mathbf{Q})$ are Zariski dense in the norm-one torus. Over $\overline{\mathbf{Q}}$ this torus is parametrized inside $E \otimes_{\mathbf{Q}} \overline{\mathbf{Q}}$ by

$$(z_1, z_1^{-1}, \dots, z_d, z_d^{-1}).$$

Here $d = [E_0 : \mathbf{Q}]$. No nonzero linear form $\sum_i (a_i z_i + b_i z_i^{-1})$ vanishes identically on this torus. Consequently the torus is contained in no proper linear subspace of E , and the same is true of its Zariski-dense rational points. This proves the claim. This is the norm-one spanning statement

used in the CM K3 argument of Ramón Marí; it is Lemma 3.2 in the arXiv version of [15]. See also [22, Remark 2.10]. $\square$

2.3. Homogeneous extraction from Markman's correspondence. We use Markman's algebraicity theorem in the following degree-by-degree form. Writing the homogeneous pieces is important because only one of them has the variance needed below.

Theorem 2.5 (Markman, specialized to $K3^{[2]}$ type). *Let F be a projective irreducible holomorphic symplectic fourfold of $K3^{[2]}$ type and let $f : H^2(F, \mathbf{Q}) \rightarrow H^2(F, \mathbf{Q})$ be a rational Hodge isometry. Then f is induced by an algebraic correspondence $C_f \in \mathrm{CH}^4(F \times F)_{\mathbf{Q}}$. More precisely, Markman's algebraic total-cohomology correspondence may be written*

$$\tilde{R}_f = \sum_{r=0}^8 R_f^{(r)}, \quad R_f^{(r)} \in \mathrm{CH}^r(F \times F)_{\mathbf{Q}},$$

and one of the following holds:

- (a) $\tilde{R}_f$ preserves degree and $(R_f^{(4)})_*|_{H^2(F)} = f$;
- (b) $\tilde{R}_f$ reverses degree and, for some $\lambda_f \in \mathbf{Q}^\times$,

$$(R_f^{(2)} \circ \Delta_{F,*} c_2(F))_*|_{H^2(F)} = \lambda_f f.$$

Thus one may take

$$C_f = R_f^{(4)} \quad \text{in case (a),} \quad C_f = \lambda_f^{-1} R_f^{(2)} \circ \Delta_{F,*} c_2(F) \quad \text{in case (b).} \quad (2.6)$$

Proof. Markman's Theorem 1.1 produces $\tilde{R}_f$ with one of the two stated grading behaviors and asserts that both the lift and the induced isometry are algebraic when F is projective [13, Theorem 1.1]. Hence its homogeneous pieces lie in the indicated rational Chow groups. A class in $\mathrm{CH}^r(F \times F)$ shifts cohomological degree by $2(r - 4)$. In the degree-preserving case, therefore, only $R_f^{(4)}$ can contribute to the map $H^2(F) \rightarrow H^2(F)$; Markman's statement that the lift restricts to f gives case (a) exactly, without an additional scalar.

In the degree-reversing case, cup product by $c_2(F)$ first sends $H^2(F)$ to $H^6(F)$ and is represented by $\Delta_{F,*} c_2(F) \in \mathrm{CH}^6(F \times F)_{\mathbf{Q}}$. Degree reversal then sends $H^6(F)$ to $H^2(F)$, so only $R_f^{(2)}$ contributes. Theorem 1.1 states that this composite is a nonzero rational multiple $\lambda_f f$. Finally,

$$\mathrm{codim}(R_f^{(2)} \circ \Delta_{F,*} c_2(F)) = 2 + 6 - 4 = 4,$$

which proves (2.6). In both cases C_f is a homogeneous rational Chow correspondence, not only a mixed-degree cohomological class. $\square$

2.4. The weighted incidence return. Let g be the Plücker class on F_X and set

$$L_{g^2} = \Delta_{F_X,*}(g^2) \in \mathrm{CH}^6(F_X \times F_X), \quad B = P \circ L_{g^2} \in \mathrm{CH}^5(F_X \times X).$$

Proposition 2.6 (g^2 -weighted return). *There is a nonzero $c \in \mathbf{Q}$ such that*

$$(B \circ A)_* = c \, \mathrm{id} \quad \text{on } V_X. \quad (2.7)$$

Equivalently,

$$B_*|_{g^\perp} = c A_*^{-1}.$$

With the normalization below, $c = -6$; only its nonvanishing will be used.

Proof. We use Hassett's Beauville–Bogomolov–Fujiki normalization; in particular, $q_{F_X}(g, g) = 6$. The polarized Fujiki relation [9, Theorem 4.7] is

$$\int_{F_X} \alpha\beta\gamma\delta = q_{F_X}(\alpha, \beta)q_{F_X}(\gamma, \delta) + q_{F_X}(\alpha, \gamma)q_{F_X}(\beta, \delta) + q_{F_X}(\alpha, \delta)q_{F_X}(\beta, \gamma).$$

Since $A_*x, A_*y \in g^\perp$, this yields

$$\int_{F_X} g^2 \cup A_*x \cup A_*y = 6q_{F_X}(A_*x, A_*y).$$

Beauville–Donagi's Abel–Jacobi map satisfies

$$q_{F_X}(A_*x, A_*y) = - \int_X x \cup y \quad (x, y \in H_{\text{prim}}^4(X, \mathbf{Q}))$$

[4, Proposition 6]; both identities with this normalization are recorded together in [10, Section 2.1 and Proposition 2.1.1]. Thus incidence adjunction gives

$$\begin{aligned} \int_X (B \circ A)_* x \cup y &= \int_{F_X} g^2 \cup A_*x \cup A_*y \\ &= 6q_{F_X}(A_*x, A_*y) \\ &= -6 \int_X x \cup y. \end{aligned}$$

The Poincaré pairing on V_X is nondegenerate, so the displayed equality for every $y \in V_X$ proves (2.7), with $c = -6$ in the chosen normalization [10, Section 2.1 and Proposition 2.1.1]. Only $c \neq 0$ is used. Since $A_* : V_X(1) \xrightarrow{\sim} g^\perp$, the equivalent formulation follows as well. $\square$

Proposition 2.7 (Fano transport of CM endomorphisms). *If E_X is CM, every element of E_X is represented by a rational Chow correspondence in $\text{CH}^4(X \times X)_{\mathbf{Q}}$.*

Proof. The polarization adjoint restricts to the positive involution on E_X ; in the CM case this is complex conjugation [24, Theorem 1.5.1]; see also [14, Section 2.1]. Thus every $u \in U_{E_X}(\mathbf{Q})$ is an isometry of T_X . By Lemma 2.2,

$$f_u = \text{id}_{\text{NS}(F_X)_{\mathbf{Q}}} \oplus A_* u A_*^{-1}$$

is a rational Hodge isometry of the full $H^2(F_X, \mathbf{Q})$. The identity on $\text{NS}(F_X)_{\mathbf{Q}}$ is essential: extension by zero would not be invertible and would not meet Markman's isometry hypothesis.

Theorem 2.5 gives $C_u \in \text{CH}^4(F_X \times F_X)_{\mathbf{Q}}$ acting as f_u on $H^2(F_X)$. In its degree-reversing alternative, formula (2.6) includes the required $c_2(F_X)$ factor. On the transcendental summands the resulting diagram is

$$\begin{array}{ccc} T_X(1) & \xrightarrow{A_*} & T_{F_X} \\ u \downarrow & & \downarrow (C_u)_* \\ T_X(1) & \xrightarrow{A_*} & T_{F_X}. \end{array} \quad (2.8)$$

By Proposition 2.6, for $x \in T_X$,

$$(B \circ C_u \circ A)_* x = c u(x).$$

The unsandwiched composition has the required codimension:

$$\text{codim}(B \circ C_u \circ A) = 5 + 4 + 3 - 4 - 4 = 4.$$

Consequently

$$\Gamma_u = c^{-1} \pi_T^X \circ B \circ C_u \circ A \circ \pi_T^X \in \text{CH}^4(X \times X)_{\mathbf{Q}} \quad (2.9)$$

acts as u on T_X . Lemma 2.4 now expresses every element of E_X as a rational linear combination of such correspondences. On the transcendental realization, the construction is the typed composite

$$T_X(1) \xrightarrow{A_*} T_{F_X} \xrightarrow{(C_u)_*} T_{F_X} \xrightarrow{c^{-1}B_*} T_X(1),$$

which is exactly u . $\square$

Remark 2.8. Bolognesi–Pedrini prove the compatible isomorphism of rational Chow motives

$$h_2^{\text{tr}}(F_X)(1) \simeq h_4^{\text{tr}}(X) = t(X)$$

using the Abel–Jacobi correspondence and a Chow-motivic inverse [5, Theorem 2.5]. The proof above instead uses the explicit g^2 -weighted cohomological return; it does not identify that motivic inverse with the transpose incidence.

3. UNITARY INVARIANTS AND THE ALL-POWER THEOREM

Lemma 3.1 (CM generation by two-factor tensors). *Let T be a simple polarized rational Hodge structure of K3 type, of weight two, whose Hodge endomorphism field E is CM. Every Hodge class in $T^{\otimes r}(r)$ is a linear combination of tensor products and permutations of two-factor Hodge tensors*

$$E = \text{End}_{\text{HS}}(T) \subset T^\vee \otimes T \simeq T^{\otimes 2}(2).$$

In particular, $(T^{\otimes r})^{\text{Hg}(T)} = 0$ when r is odd.

Proof. Let E_0 be the maximal totally real subfield of E . Zarhin identifies the Hodge group with the restriction of scalars from E_0 of the unitary group of the E/E_0 -Hermitian polarization [24, Theorem 2.3.1]; see also [14, Section 2.1]. For each embedding $\sigma : E_0 \hookrightarrow \mathbf{R}$, choose the two extensions $\rho, \bar{\rho} : E \hookrightarrow \mathbf{C}$ and put $W_\rho = T \otimes_{E, \rho} \mathbf{C}$. The Hermitian form identifies $W_{\bar{\rho}}$ with $W_\rho^\vee$. Consequently

$$\text{Hg}(T)_{\mathbf{C}} \simeq \prod_{\sigma : E_0 \hookrightarrow \mathbf{R}} \text{GL}(W_\rho), \quad T_{\mathbf{C}} \simeq \bigoplus_{\sigma} (W_\rho \oplus W_\rho^\vee).$$

For each general-linear factor, the first fundamental theorem says that a mixed tensor is invariant only when its standard and dual occurrences can be paired, and that all invariants are spanned by complete evaluation contractions. Invariants of the product group are products of the colored contractions, one color for each ρ .

We now identify those complex contractions inside the complexification of a rational space. The polarization $q : T \otimes T \rightarrow \mathbf{Q}(-2)$ identifies $T^\vee$ with $T(2)$. For $e \in E$, let $t_e \in T^{\otimes 2}(2)$ be the image of $e \in T^\vee \otimes T$ under this identification. Equivalently, up to the fixed choice of which tensor factor is dualized, t_e is characterized by the bilinear form

$$q_e(x, y) = q(ex, y). \quad (3.1)$$

Changing the action convention replaces e by its polarization adjoint; the adjoint preserves E , so it does not alter the generated rational space.

Let $D_r \subseteq (T^{\otimes r})^{\text{Hg}(T)}$ be the rational span of tensor products and permutations of the t_e . Since $\text{Hg}(T)$ is reductive, invariants commute with extension from $\mathbf{Q}$ to $\mathbf{C}$. On the other hand,

$$E \otimes_{\mathbf{Q}} \mathbf{C} = \bigoplus_{\rho : E \hookrightarrow \mathbf{C}} \mathbf{C} e_\rho,$$

where e_ρ is the primitive idempotent projecting to the ρ -eigenspace. These idempotents need not lie in E itself, but they do lie in the complexification of the rational vector space E . If ρ lies over σ , the tensor t_{e_ρ} is precisely the colored evaluation between W_ρ and $W_{\bar{\rho}} = W_\rho^\vee$. Thus

$D_r \otimes_{\mathbf{Q}} \mathbf{C}$ contains every complete colored contraction. The reverse inclusion is tautological, and the product first fundamental theorem therefore gives

$$D_r \otimes_{\mathbf{Q}} \mathbf{C} = (T_{\mathbf{C}}^{\otimes r})^{\mathrm{Hg}(T)\mathbf{C}} = ((T^{\otimes r})^{\mathrm{Hg}(T)}) \otimes_{\mathbf{Q}} \mathbf{C}.$$

Faithful flatness yields $D_r = (T^{\otimes r})^{\mathrm{Hg}(T)}$. If r is odd there is no complete pairing, so this space is zero. This is the invariant-theoretic argument in the CM part of Ramón Mari's Proposition 5.2 [15], reproved by Varesco [21, Theorem 2.2]. The false totally real extension of the former proposition is irrelevant here; see [21, Remark 2.17]. Crucially, the complex group is a product of full general linear groups, not special linear groups; consequently its invariant algebra has no additional determinant generator. $\square$

3.1. From invariant tensors to cycles on every power.

Proposition 3.2 (Tensor-to-cycle export). *Let X be a smooth cubic fourfold for which E_X is CM. Suppose every $e \in E_X$ is induced on T_X by a class in $\mathrm{CH}^4(X \times X)_{\mathbf{Q}}$. Then $\mathrm{HC}_{\mathbf{Q}}(X^m)$ holds in every codimension for every $m \geq 1$.*

Proof. By replacing a correspondence Γ_e with $\pi_T^X \circ \Gamma_e \circ \pi_T^X$, we may assume that its Betti class is supported on the $T_X \otimes T_X$ Künneth summand. Lemma 2.1 gives a graded decomposition

$$H^*(X, \mathbf{Q}) = T_X \oplus A_X, \quad A_X = \mathrm{im}((\pi_{\mathrm{alg}}^X)_*), \quad (3.2)$$

induced by Chow idempotents. The space A_X has a homogeneous algebraic basis: by the Lefschetz hyperplane theorem and Poincaré duality, the odd cohomology vanishes and the nonmiddle cohomology is generated by $1, h, h^3, h^4$; adding h^2 and the chosen basis of N_X gives the stated basis. The complementary projector π_{alg}^X is not a single pure Künneth projector; rather, it is the sum of these homogeneous algebraic summands.

For $I \subseteq \{1, \dots, m\}$ put

$$\Pi_I = \mathrm{mathop{\boxtimes}}_{j \in I} \pi_T^X \boxtimes \boxtimes_{j \notin I} \pi_{\mathrm{alg}}^X \in \mathrm{CH}^{4m}(X^m \times X^m)_{\mathbf{Q}}. \quad (3.3)$$

The realizations of the Π_I are mutually orthogonal and sum to the identity. Let α be a rational Hodge class on X^m and fix I , with $r = |I|$. Expand the A_X positions of $(\Pi_I)_* \alpha$ in a homogeneous algebraic basis and extract their coefficients using the dual algebraic basis. This writes $(\Pi_I)_* \alpha$ as a finite sum

$$\sum_{\nu} t_{\nu} \otimes b_{\nu}, \quad (3.4)$$

where every b_{ν} is an algebraic tensor on the complementary positions and

$$t_{\nu} \in (T_X^{\otimes r}(2r))^{\mathrm{Hg}(T_X)}.$$

Indeed, all T_X positions have cohomological degree four, so a Hodge tensor in those positions has type $(2r, 2r)$. The basis and coefficient-extraction steps in (3.4) are induced by algebraic cycles.

Apply Lemma 3.1 to the weight-two Hodge structure $T_X(1)$. If r is odd, the invariant space above is zero. If r is even, every t_{ν} is a rational linear combination of complete pairings whose edges are labelled by elements of E_X . For a perfect matching $\mathcal{P}$ of the r labelled positions and labels $e_{ij} \in E_X$, define

$$Z(\mathcal{P}, (e_{ij})) = \prod_{(i,j) \in \mathcal{P}} p_{ij}^* \Gamma_{e_{ij}} \in \mathrm{CH}^{2r}(X^r)_{\mathbf{Q}}. \quad (3.5)$$

There are $r/2$ factors in the product and each has codimension four, so the displayed codimension is $2r$. Under the Künneth identification, its $T_X^{\otimes r}$ component is exactly the corresponding

complete contraction. If the opposite correspondence convention is used, the edge label is replaced by its polarization adjoint; the adjoint still lies in E_X .

Exterior product with the algebraic cycles representing the b_ν , followed by the required factor permutations, therefore gives a Chow representative of every term in (3.4). Summing over I and ν gives a Chow representative of α . This proves the proposition in every degree and codimension. $\square$

Proof of Theorem A. Proposition 2.7 supplies the required correspondence Γ_e for every $e \in E_X$. Proposition 3.2 then proves the all-power assertion. The first assertion of the theorem is exactly Proposition 2.7. $\square$

4. CYCLIC CUBIC FOURFOLDS AND THE GRAPH-PFAFFIAN

4.1. Eisenstein primitive cohomology. The quotient map

$$q : Y \longrightarrow Y/\langle \tau \rangle \simeq \mathbf{P}^4$$

is the cyclic triple cover branched along the cubic threefold $\{F = 0\} \subset \mathbf{P}^4$. Rational transfer gives

$$q_* q^* = 3 \text{ id}, \quad q^* q_* = 1 + \tau^* + (\tau^*)^2.$$

These identities remain valid for the finite branched quotient: after an equivariant triangulation, averaging over the finite deck group identifies rational cochains on the quotient with invariant rational cochains upstairs. Hence the invariant cohomology is the pullback of the cohomology of the quotient. In middle degree it is $q^* H^4(\mathbf{P}^4, \mathbf{Q}) = \mathbf{Q} h^2$, and therefore

$$V^\tau = 0. \tag{4.1}$$

This quotient description, the rank-11 Eisenstein module structure, and the signature (10, 1) Hermitian form are recorded by Allcock–Carlson–Toledo [1, Section 2]; see also van Geemen–Izadi [20, Section 2.1] and Laterveer [11, Remark 3.4].

Put $T = \tau^*|_V$. Since $T^3 = 1$ and $V^T = 0$,

$$T^2 + T + 1 = 0. \tag{4.2}$$

Thus V is a vector space of dimension 11 over $K = \mathbf{Q}(\zeta_3)$. At one complex embedding of K , the eigenspace has Hodge numbers

$$\dim V_{\zeta_3}^{3,1} = 1, \quad \dim V_{\zeta_3}^{2,2} = 10, \tag{4.3}$$

and the conjugate eigenspace carries the conjugate Hodge types [1, Section 2].

The distinction between $\mathbf{Q}$ - and K -determinants will be useful below. The one-dimensional K -space $\bigwedge_K^{11} V$ is only SU -invariant and has nonzero Hodge-circle character, so neither of its complex embedding lines is a Hodge line. Their tensor product under the two embeddings is the rational norm line $\bigwedge_{\mathbf{Q}}^{22} V$; it is U -invariant and will be represented by the graph-Pfaffian cycle.

Proposition 4.1 (The every-fiber CM property). *For every smooth cyclic cubic fourfold Y , the field*

$$E_Y = \text{End}_{\text{HS}}(T_Y)$$

is CM. Consequently $\text{HC}_{\mathbf{Q}}(Y^m)$ holds in every codimension for every $m \geq 1$.

Proof. Lemma 2.3, applied to Y , shows that $T_Y(1)$ is simple and that E_Y is a number field. The deck transformation preserves N_Y and Q_Y , and therefore preserves its orthogonal complement $T_Y \subseteq V$. Moreover $H^{3,1}(Y) \subset (T_Y)_{\mathbf{C}}$, so $T_Y \neq 0$. Equation (4.1) gives $T_Y^\tau = 0$; hence the

restriction of τ to T_Y is not the identity. Since $\tau^3 = 1$, its minimal polynomial is exactly $t^2 + t + 1$. Hence

$$K = \mathbf{Q}[\tau] \simeq \mathbf{Q}(\zeta_3) \subseteq E_Y.$$

Zarhin's dichotomy says that E_Y is either totally real or CM [24, Theorem 1.5.1]. A totally real field cannot contain the imaginary quadratic field K , so E_Y is CM. The all-power assertion is now Theorem A. $\square$

4.2. The graph-Pfaffian cycle. By the correspondence convention fixed in Section 2, $(\Gamma_g)_* = g_* = (g^{-1})^*$. Therefore the action of β on V is

$$S = T^{-1} - T. \quad (4.4)$$

The opposite graph convention changes S and the final eleventh wedge by a sign and changes no nonvanishing statement.

Lemma 4.2. *The operator S is invertible, satisfies $S^2 = -3$, and is skew-adjoint for the primitive intersection form Q_Y .*

Proof. Equation (4.2) and $T^{-1} = T^2$ give

$$S^2 = (T^2 - T)^2 = T^4 - 2T^3 + T^2 = T + T^2 - 2 = -3.$$

The deck transformation preserves Q_Y , hence $T^\dagger = T^{-1}$ and $S^\dagger = T - T^{-1} = -S$. $\square$

The graph difference acts by zero on every ambient Tate summand $\mathbf{Q}h^i \subset H^*(Y, \mathbf{Q})$, because τ fixes h . Thus its cohomology class is already purely primitive. We nevertheless retain the projectors in (1.2), making the primitive restriction in the identity $S^2 = -3$ explicit.

Use Q_Y to identify V with $V^\vee$. We fix the tensor–endomorphism identification

$$V \otimes V \longrightarrow \text{End}(V), \quad a \otimes b \longmapsto (x \longmapsto Q_Y(x, a)b). \quad (4.5)$$

Under (4.5), the tensor $b = \text{cl}(\beta)$ corresponds to S . Since S is skew-adjoint, b lies in $\bigwedge^2 V$; since S is invertible, it is a nondegenerate alternating bivector. We call the alternating top exterior power obtained from this graph-difference tensor the *graph-Pfaffian*.

Lemma 4.3 (Pfaffian identity). *If $b \in \bigwedge^2 V$ is a nondegenerate alternating bivector on a 22-dimensional vector space, then*

$$P_{\text{alt}}(b^{\otimes 11}) \neq 0 \quad \text{in} \quad \bigwedge^{22} V.$$

In a basis with skew coefficient matrix B , the coefficient is a nonzero normalization multiple of $\text{Pf}(B)$.

Proof. After extending scalars, choose a symplectic basis in which $b = \sum_{i=1}^{11} u_i \wedge v_i$. Its top exterior power is

$$b^{\wedge 11} = 11! u_1 \wedge v_1 \wedge \cdots \wedge u_{11} \wedge v_{11},$$

which is nonzero. The tensor alternator differs from the exterior product only by a nonzero rational normalization. Equivalently, direct expansion gives a nonzero multiple of $\text{Pf}(B)$, whose square is $\det(B) \neq 0$. $\square$

Applying Lemma 4.3 to $b = \text{cl}(\beta)$ proves the explicit graph-Pfaffian clause of Theorem B. Every operation in (1.3) is a Chow operation. Thus z_τ is defined directly from algebraic correspondences on Y . Here $\beta^{\boxtimes 11}$ is placed on the ordered pairs

$$(1, 2), (3, 4), \dots, (21, 22)$$

of factors of Y^{22} . Under the Künneth identification its primitive class is $b^{\otimes 11}$, and the normalized factor alternator in (1.3) is exactly the projection to $\bigwedge^{22} V$. This fixes both the tensor-block ordering and the normalization independently of any choice of basis of V . Since the Hodge numbers of V are $(1, 20, 1)$ in bidegrees $(3, 1), (2, 2), (1, 3)$, respectively, its top exterior power has bidegree

$$(3 + 20 \cdot 2 + 1, 1 + 20 \cdot 2 + 3) = (44, 44).$$

Because $\dim_{\mathbf{Q}} V = 22$, the space $\bigwedge_{\mathbf{Q}}^{22} V$ is one-dimensional. Thus the nonzero rational class produced by the Pfaffian calculation is a Hodge class of codimension 44 on Y^{22} and spans that line. Together with Proposition 4.1, this completes the proof of Theorem B.

Remark 4.4. The nonvanishing of z_τ is not used to prove the all-power statement. That statement already follows from Proposition 4.1 and Theorem A. The graph-Pfaffian is instead a basis-free rational Chow representative of the primitive orientation line. In the generic unitary calculation below, that line is itself generated by the two-factor contractions, so no additional special-unitary generator is being inserted into the proof.

5. KUZNETSOV-COMPONENT MODULI AND OG10 APPLICATIONS

We work in the contravariant category of rational Chow motives, with

$$\mathrm{Hom}(h(Z), h(W)(s)) = \mathrm{CH}^{\dim Z + s}(Z \times W)_{\mathbf{Q}}. \quad (5.1)$$

The geometric applications follow from a formal export principle and the split Chow-motive retractions proved by Fu–Laterveer–Vial. Splitness is essential: the incoming correspondence sends a Hodge class to powers of the fixed cubic, while the outgoing correspondence returns the original class because the composite is the diagonal. A nonzero rational scalar multiple would likewise suffice after normalization; a one-way inclusion or merely cohomological containment would not.

Lemma 5.1 (Split-motive export). *Let X be a smooth projective variety satisfying the rational Hodge conjecture for every power. Suppose that, for $a = 1, \dots, s$, the rational Chow motive $h(Z_a)$ is a direct summand of a finite direct sum of Tate twists of motives $h(X^{r_{a,j}})$. Then the rational Hodge conjecture holds for*

$$X^b \times \prod_{a=1}^s Z_a$$

for every $b \geq 0$. The conclusion is preserved when a factor Z_a is replaced by a smooth projective irreducible-symplectic variety birational to it.

Proof. Choose split Chow correspondences

$$h(Z_a) \xrightarrow{i_a} \bigoplus_j h(X^{r_{a,j}})(t_{a,j}) \xrightarrow{p_a} h(Z_a), \quad p_a i_a = \mathrm{id}_{h(Z_a)}.$$

Writing $Z = X^b \times \prod_a Z_a$, tensoring the displayed maps gives

$$I = \mathrm{id}_{h(X^b)} \otimes \bigotimes_a i_a, \quad R = \mathrm{id}_{h(X^b)} \otimes \bigotimes_a p_a, \quad R \circ I = \mathrm{id}_{h(Z)}. \quad (5.2)$$

The middle motive is a finite direct sum of terms

$$h(X^{b + \sum_a r_{a,j_a}}) \left(\sum_a t_{a,j_a} \right).$$

In explicit codimension conventions, a component through $h(X^n)(s)$, with $d_a = \dim Z_a$ and $e = \dim X$, is represented by

$$\gamma_a \in \mathrm{CH}^{d_a+s}(Z_a \times X^n)_{\mathbf{Q}}, \quad \delta_a \in \mathrm{CH}^{en-s}(X^n \times Z_a)_{\mathbf{Q}}.$$

For a rational Hodge class α on Z , the class $I_*\alpha$ is a finite direct sum of Hodge classes on powers of X , with Tate twists. Those classes are algebraic by hypothesis: a Tate twist only changes the codimension in which the cycle class is sought. The outgoing Chow correspondence returns the resulting algebraic cycles to Z , and $R_*I_*\alpha = \alpha$ by (5.2). Thus the argument uses an actual split in the category of rational Chow motives, not merely a split of Hodge structures. Exterior products add the twists, and the required middle-factor permutation commutes with the full exterior-product identity. Riess's mutually inverse Chow correspondences give a Chow-motive isomorphism between birational projective irreducible-symplectic varieties [16, Theorem 3.2 and its proof of invertibility], so the last assertion follows. $\square$

Proposition 5.2 (Published motivic retractions). *Let X be a smooth cubic fourfold.*

- (i) *If v is primitive and $M = M_\sigma(\mathrm{Ku}(X), v)$ is a nonempty smooth projective v -generic moduli space of dimension $2m$, with $\sigma \in \mathrm{Stab}^\dagger(\mathrm{Ku}(X))$, then $h(M)$ is a direct summand of a finite direct sum of Tate twists of $h(X^m)$.*
- (ii) *If $v = 2v_0$ with v_0 primitive and $v_0^2 = 2$, then every projective hyperkähler crepant resolution of a nonempty moduli space $M_\sigma(\mathrm{Ku}(X), v)$, with σ v -generic, that lies in the scope of [8, Theorem 1.3], has Chow motive a direct summand of a finite direct sum of Tate twists of $h(X^5)$.*

Proof. Part (i) is exactly [8, Theorem 1.1]. Its proof uses the complete universal-Ext diagonal; Floccari–Fu–Zhang's factorization is an underlying input [6, Proposition 5.2]. The authors' erratum does not modify that proposition [7]. Part (ii) is [8, Theorem 1.3]. For the primitive moduli in part (i), existence and smooth-projectivity results are supplied by Bayer–Lahoz–Macrì–Nuer–Perry–Stellari [3, Theorem 29.2]. Li–Pertusi–Zhao construct the projective symplectic resolution in [12, Theorem 1.1; see also Theorem 3.1], and identify the relevant birational twisted-intermediate-Jacobian model in [12, Theorem 1.3]. This last passage is also recorded in [8, Remark 1.4] and enters through Riess's birational invariance, not through an unstated motive comparison. $\square$

Apply Lemma 5.1 to the cubic X and use Theorem A, with the retractions of Proposition 5.2. This proves Theorem C.

6. SCOPE OF THE METHOD

The condition that E_X is a CM field is an endomorphism-field condition; it does not assert that the Mumford–Tate group of T_X is a torus. In general Zarhin's Hodge group is a positive-dimensional unitary group. This is precisely why higher tensor invariants occur, and why the passage from algebraic endomorphisms to all powers requires the unitary first fundamental theorem.

The CM hypothesis is the exact boundary of the argument given here. For a totally real endomorphism field, Zarhin's Hodge group is orthogonal. Pair contractions generate the invariants of the full orthogonal group, but the actual special-orthogonal Hodge group also has an orientation tensor in the degree equal to the dimension over the endomorphism field. That tensor is not generated by two-factor endomorphism tensors. The present proof therefore makes no assertion in the totally real case.

Every application uses one source cubic: each motivic factor retracts from a power of the same fixed X . Products attached to distinct cubics would require control of Hodge tensors between

their transcendental structures and are not covered. Likewise, the arguments use ordinary rational Chow groups over $\mathbf{C}$; they do not address integral Hodge classes, descent of cycles to a prescribed arithmetic field, the Tate conjecture, or the general Hodge conjecture. For a cyclic equation defined over a smaller field, conjugation may exchange τ and τ^{-1} and hence negate the chosen graph-Pfaffian cycle, without affecting its complex algebraicity or nonvanishing.

Use of AI tools. OpenAI language models, including Codex, were used extensively in mathematical exploration, drafting, source identification, and the development of computational checks. They are not authors. The named author assumes full responsibility for all mathematical statements, proofs, references, and wording. No proof depends on an unprinted computer-generated calculation; the finite computations used during development are auxiliary checks of formulas that are proved in the text.

APPENDIX A. GENERIC CYCLIC HODGE GROUP AND EXPLICIT GENERATORS

Theorem B is already proved for every smooth member. We now identify the generic group in order to obtain the explicit two-generator description below.

Proposition A.1 (Generic cyclic Hodge group). *At a Hodge-generic point of the smooth cyclic-cubic family,*

$$\mathrm{Hg}(V) = \mathrm{U}_K(V, H), \quad N_Y = 0, \quad \mathrm{End}_{\mathrm{HS}}(T_Y) = K.$$

Proof. Let U be the Zariski-open locus of smooth cubic forms in $\mathbf{P}H^0(\mathbf{P}^4, \mathcal{O}(3))$, and let $\mathbb{V}$ be the primitive variation of Hodge structure of the associated cyclic cubics. Here Hodge-generic means outside the countable union of proper Hodge loci in U . Allcock–Carlson–Toledo prove that the integral linear monodromy consists of the Eisenstein-lattice isometries whose K -determinant is a cube root of unity [1, Theorem 7.1 in the published version; Theorem 8.1 in the arXiv version]. Its determinant-one subgroup is an arithmetic lattice in $\mathrm{SU}(10, 1)$. Borel density therefore gives

$$G_{\mathrm{mon}}^{\circ} = \mathrm{SU}_K(V, H). \quad (\mathrm{A.1})$$

Thus the connected monodromy is the special unitary group.

Let $G = \mathrm{Hg}(V)$ at a Hodge-generic point; by convention G is the connected Hodge group. The K -action and polarization are Hodge tensors, so

$$G \subseteq \mathrm{U}_K(V, H). \quad (\mathrm{A.2})$$

The theorem of the fixed part in Mumford–Tate form places connected algebraic monodromy inside the derived generic Mumford–Tate group; in the form used here this is Andre [2, Section 5, Theorem 1]. Since the derived group of $\mathrm{U}_K(V, H)$ is $\mathrm{SU}_K(V, H)$, the two inclusions give

$$\mathrm{SU}_K(V, H) \subseteq G, \quad G^{\mathrm{der}} = \mathrm{SU}_K(V, H). \quad (\mathrm{A.3})$$

The remaining central torus is detected by the Hodge circle. After a Tate twist, the ζ_3 eigenspace has one line of type $(1, -1)$ and ten dimensions of type $(0, 0)$. Its K -determinant on the Hodge circle is $z^{\pm 2}$, and is therefore nonconstant. Since

$$1 \longrightarrow \mathrm{SU}_K(V, H) \longrightarrow \mathrm{U}_K(V, H) \xrightarrow{\det_K} \mathrm{Res}_{K/\mathbf{Q}}^1 \mathbf{G}_m \longrightarrow 1$$

has a one-dimensional quotient, the connected image of G in this quotient is either trivial or the whole torus. Its restriction to the Hodge circle is nontrivial, so it is the whole torus. Equations (A.2) and (A.3) now give

$$\mathrm{Hg}(V) = \mathrm{U}_K(V, H). \quad (\mathrm{A.4})$$

The standard K -representation of the full unitary group has centralizer K and has no nonzero invariant vector. Therefore the centralizer of $\mathrm{Hg}(V)$ in $\mathrm{End}_{\mathbf{Q}}(V)$ is K , and V has no nonzero

rational Hodge classes of type $(2, 2)$. Thus at a Hodge-generic point $N_Y = 0$, $T_Y = V$, and $\text{End}_{\text{HS}}(T_Y) = K$. $\square$

A.1. Explicit deck generators at a generic cyclic point. At a Hodge-generic point, Proposition A.1 gives $T_Y = V$ and $E_Y = K = \mathbf{Q}(\zeta_3)$. Put $W = T_Y(2)$ and $A = T - T^{-1} \in K$. The elements $1, A$ form a rational basis of K . Specializing Lemma 3.1 to this basis shows that $(W^{\otimes r})^{\cup_K(W, H)}$ vanishes for odd r and, for even r , is spanned by complete pair contractions whose edges are labelled by 1 or A . Indeed, after complexification $W_{\mathbf{C}} = E \oplus E^{\vee}$ and the two eigenvalues $\pm\sqrt{-3}$ of A separate the two colored evaluation pairings.

Above the stable range the spanning contractions satisfy the expected determinantal relations. No independence assertion is needed. The lemma also explains why the graph Pfaffian is not an additional exceptional unitary generator: the rational orientation is already a polynomial in the two pair contractions.

Now specialize again to a cyclic Y with $E_Y = K$. Put

$$\beta_T = \pi_T^Y \circ (\Gamma_{\tau} - \Gamma_{\tau^{-1}}) \circ \pi_T^Y.$$

The tensor for the identity in K is represented by π_T^Y , and the second K generator is represented, up to the harmless graph-convention sign, by β_T . Lemma 3.1 gives the stronger explicit description: every contraction is the class of an external product of pullbacks of π_T^Y and β_T along pair projections

$$p_{ij} : Y^m \longrightarrow Y \times Y.$$

For a Hodge-generic cyclic cubic, $N_Y = 0$ and $\pi_T^Y = \pi_V$; then $\beta_T = \beta$. Combining this primitive tensor description with the algebraic ambient Künneth factors gives the following.

Corollary A.2 (Generic cyclic generators). *For a Hodge-generic cyclic cubic Y , one has $N_Y = 0$ and $E_Y = \mathbf{Q}(\zeta_3)$. For every $m \geq 1$, the rational Hodge classes on Y^m are spanned by products of pullbacks of hyperplane powers and of the two algebraic pair tensors*

$$\pi_V, \quad \beta \quad \text{in } \text{CH}^4(Y \times Y)_{\mathbf{Q}}.$$

Proof. Proposition A.1 gives $N_Y = 0$ and $E_Y = K$. Lemma 3.1 handles every primitive tensor slot, while the algebraic Künneth decomposition handles all ambient slots. Pulling the two pair cycles back along pair projections realizes the resulting contractions in rational Chow. $\square$

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